

1 Critical evaluation of the modified Gompertz response
2 function as a parametric template for
3 treatment-response trajectories in N-of-1 trial
4 simulation

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69 results inform protocol-design decisions, and we propose a default sensitivity grid
70 that the framework can run alongside its primary cells.

71 2 Introduction

72 2.1 A claim about response-curve choice in N-of-1 trial simulation

73 We begin by setting out two coupled methodological claims that this paper devel-
74 ops.

75 **First**, the modified Gompertz function used as the parametric template for
76 treatment-response trajectories in aggregated N-of-1 trial simulation (specifically,
77 in the pmsimstats simulation framework that operationalises the Hendrickson et
78 al. design family¹) is a defensible default for the saturating-shape regime that
79 characterises most pharmacodynamic and chronic-condition response curves over
80 the typical trial timescale. The function is well-established in tumour growth
81 dynamics^{2,3}, microbial growth modelling⁴, and the broader unified- Richards
82 growth-curve family⁵⁻⁷; its three-parameter form is flexible enough to capture
83 the asymmetric saturation profiles that linear extrapolation, exponential decay,
84 or symmetric logistic models cannot.

85 **Second**, family-misspecification sensitivity is essentially absent from the pub-
86 lished N-of-1 methodology literature, and **should be reported routinely**, not
87 assumed away. The methodological N-of-1 corpus^{1,8-10} is silent on parametric
88 trajectory shape: papers either inherit a chosen family from the upstream liter-
89 ature without comparing alternatives, or specify a linear or AR(1) form that is
90 too coarse to capture the saturation that real trajectories exhibit. The oncology-
91 modelling literature has set the precedent for what a proper family-comparison
92 sensitivity analysis looks like^{6,11} and the pharmacology literature has converged
93 on Hill / sigmoid-Emax / indirect-response models for related questions^{12,13}. The
94 N-of-1 literature has not imported this comparative apparatus. The present paper
95 supplies it.

96 The two claims are coupled because the case for a chosen family is incomplete
97 without the comparison. We do not argue that Gompertz is the *correct* family in
98 any strict sense; rather, we argue that it is a reasonable choice within the unified-
99 Richards family of saturating curves, and that the inferential consequences of
100 choosing it (versus the symmetric logistic, the von Bertalanffy growth curve¹⁴, or
101 a piecewise-linear breakpoint model) should be quantified rather than asserted.
102 The paper reports a simulation study that does so for the standard pmsimstats
103 linear-mixed analysis at trial-relevant sample sizes.

104 2.2 Why the question matters

105 We note three reasons that motivate the family-misspecification question for N-
106 of-1 simulation specifically.

107 The chosen family loads into the data before any analysis-side question

108 **is asked.** A power simulation that draws every trajectory from a Gompertz curve
 109 and then analyses each under a standard linear mixed-effects model with treat-
 110 ment indicator and AR(1) residual correlation cannot detect the inferential cost
 111 of the family choice itself. Robustness of the inference to the family is a property
 112 of the joint DGP-and-analysis specification; only a comparative simulation that
 113 varies the DGP family while holding the analysis fixed can quantify it. This is the
 114 same logic that underpins analysis-model misspecification studies in the broader
 115 clinical-trial methodology literature¹⁵, applied one level down in the simulation
 116 hierarchy.

117 **The function family has known identifiability fragility in short noisy**
 118 **trajectories.** Joint estimation of the three Gompertz parameters (m, d, r) in
 119 $y = m \exp(-d \exp(-rt))$ is poorly conditioned: the asymptote and rate param-
 120 eters correlate strongly, and the shape parameter is essentially unidentifiable when
 121 only a few timepoints lie in the inflection region.⁶ and⁷ propose unified-Richards
 122 reparameterisations in which the parameters become biologically interpretable
 123 (maximum relative growth rate, age at inflection, value at inflection, asymptote)
 124 with lower correlation;¹⁶ develops the formal *sloppiness* diagnostic that quanti-
 125 fies this directly as a property of the parameter space.¹¹ fitted eight tumour-
 126 growth models (exponential, exponential-linear, power-law, Gompertz, logistic,
 127 generalised logistic, von Bertalanffy, dynamic-carrying-capacity) to two in vivo
 128 systems and ranked them on goodness-of-fit, parametric identifiability, and pre-
 129 dictive power; the power-law model was the most identifiable in lung data while
 130 Gompertz was best for breast. The same kind of comparative diagnosis has not
 131 been performed for short N-of-1 trajectories.

132 **The default in N-of-1 simulation reflects historical borrowing rather**
 133 **than empirical optimality.** The Gompertz form entered the pmsimstats frame-
 134 work via the broader growth- curve tradition, where its use traces to Laird's
 135 tumour dynamics work² and to Zwietering's microbial growth reformulations⁴.
 136 The pharmacology literature, addressing related questions about dose-response
 137 and time-course of pharmacological action, defaults to the Hill / sigmoid-Emax
 138 family^{12,17} and to Jusko's indirect-response models¹³. These two traditions have
 139 largely not spoken to each other; the N-of-1 simulation literature has inherited
 140 the growth-biology default without an explicit defence against the pharmacology
 141 default. Either default may be adequate for the saturation regime, but the inher-
 142 itance is not itself evidence.

143 2.3 What the published N-of-1 literature does and does not do

144 A review of the recent N-of-1 methodology literature confirms, to the best of our
 145 knowledge, a silence on this question.

146 The CONSORT extension for N-of-1 trials¹⁰ codifies reporting standards but
 147 does not address parametric trajectory shape. The foundational programmatic
 148 statements^{8,9} emphasise design and analysis broadly without committing to a
 149 parametric response form. The Hendrickson et al. framework¹ introduces the

modified Gompertz template into the simulation engine for biomarker-validation power analyses without comparing it against alternatives. Recent applied papers in chronic-condition N-of-1 (e.g., the prazosin-PTSD reference work building on¹⁸) inherit the framework without examining the choice.

In adjacent literatures the comparative apparatus exists. The oncology-modelling community routinely fits multiple growth families and compares on identifiability and predictive performance¹¹. The pharmacometrics community produces explicit sensitivity studies for sigmoid-Emax and indirect-response models^{19,20}. The latent-class growth-trajectory literature treats response-curve heterogeneity through mixture modelling²¹. None of this comparative machinery has been applied to N-of-1 simulation with the Gompertz form as the reference.

The paper's contribution is to import that machinery, specifically: a four-family comparison (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint) crossed against the standard pmsimstats linear-mixed analysis, with results reported in the Morris-White-Crowther¹⁵ ADEMP framework already adopted as the project-wide simulation-reporting standard.

2.4 What this paper contributes

We make four contributions, each of which is addressed in a section below.

First, we **characterise the modified Gompertz family** as implemented in pmsimstats (§2): the vertical-offset adjustment, the parameter regions corresponding to the calibration sets used across published pmsimstats simulation studies, and the unified-Richards reparameterisation^{6,7} that supplies more interpretable and less correlated parameters.

Second, we **survey three alternative response-curve families** (§3): the symmetric logistic²², the hyperbolic-tangent (formally a rescaled symmetric logistic), and a piecewise-linear breakpoint model. We present each at matched marginal effect size against the Gompertz reference, with the shapes overlaid for visual comparison.

Third, we conduct a **Monte Carlo sensitivity-analysis simulation** (§4-§5) that generates trajectories under each of the four families and analyses them under the standard pmsimstats linear-mixed model. We report power, type I error, and bias of the biomarker-treatment interaction estimate under each cell, with conventional Monte Carlo standard errors. The simulation borrows the comparative methodology of¹¹ (eight-model tumour-growth comparison) and the sloppiness diagnostic of¹⁶ for small-sample identifiability assessment.

Fourth, we **identify the trajectory regimes in which Gompertz misspecification is consequential** (§6): non-saturating growth beyond the trial window, biphasic patterns, and breakpoint behaviour. For these regimes, the standard N-of-1 inference is materially affected by the family choice, and we recommend specific sensitivity-grid extensions that pmsimstats users should run alongside their primary cells.

191 The full mathematical and historical context, including citations that did not
192 survive editing into the manuscript proper, is preserved in the companion technical
193 document at `docs/25-gompertz-model-evaluation.md`.

194 3 The modified Gompertz function in `pmsimstats`

195 The Gompertz curve originates in Gompertz’s 1825 law of human mortality²³ and
196 was first put to use as a general growth curve by²⁴; the `pmsimstats` framework
197 adopts the modified, monotone-saturating form of that curve. The framework
198 parameterises each of the three latent response components (biological response
199 BR, placebo-belief response PB, and time-variant natural-history component TV)
200 with a modified Gompertz function

$$y(t) = \text{maxr} \cdot \exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t)), \quad (1)$$

201 with vertical-offset adjustment so that $y(0) = 0$. The three parameters are
202 `maxr` (asymptotic maximum response, matching the on-drug or on-natural-
203 history saturation level), `rate` (controls how quickly the response approaches
204 its asymptote), and `disp` (controls the curvature near $t = 0$). The Gom-
205 pertz form is monotone-saturating and asymmetric: the inflection occurs at
206 $t^* = \log(\text{disp})/\text{rate}$, with the rise being slower before the inflection and faster
207 after. The parameterisation used at the prazosin/PTSD reference cell sets
208 $\text{maxr}_{BR} \approx 4$ nightmares per week, $\text{rate}_{BR} \approx 0.5$ per week, and $\text{disp}_{BR} \approx 4$ for
209 biological response; analogous parameter sets for PB and TV are documented in
210 `docs/25-gompertz-model-evaluation.md` and `R/utilities.R`.

211 4 Alternative response-curve families

212 Three alternative families serve as sensitivity-analysis comparators. Each is pa-
213 rameterised so that the asymptotic maximum and a single-anchor calibration point
214 (the response level at $t = 5$ weeks, calibrated to the Gompertz family at the
215 same anchor) are matched across families. The families differ in shape envelope:
216 smoothness, symmetry around the inflection, and curvature.

217 4.1 Symmetric logistic

218 The logistic function

$$y(t) = \frac{\text{maxr}}{1 + \exp(-\text{rate} \cdot (t - t_0))} - \text{offset} \quad (2)$$

219 is monotone-saturating and *symmetric* around its inflection at $t = t_0$. A vertical
220 offset is subtracted so that $y(0) = 0$. At matched `maxr` and matched response level
221 at the calibration anchor, the logistic family rises faster than Gompertz before
222 the inflection and slower after, producing a visibly different shape envelope while

(Study 0) that matches all four to a common marginal effect size, the four-family power-and-bias factorial (Study 1, 144 cells), and the sloppiness eigenvalue diagnostic (Study 2) before the production runs.

For the present submission the simulation is scoped to a 16-cell focused factorial: four trajectory families crossed with two DGP architectures (mean moderation and MVN differential correlation, following the companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/`) crossed with two biomarker effect-size levels ($c_{bm} \in \{0, 0.45\}$). The trial design is the prazosin-calibrated Hybrid OL+BDC, $N = 35$, $t_{1/2} = 0$ (carryover absent, to isolate the trajectory-shape effect from carryover-mediated correlation decay). Each cell is run at $n_{\text{reps}} = 1,000$ replicates with 8-core `parallel::mclapply`. The full 144-cell grid (sample sizes, effect sizes, DGP-side carryover) anticipated in the original abstract is deferred to a follow-up; the focused 16-cell factorial answers the central scientific question (does trajectory family choice matter for the bm:Dcb Wald test?) at MCSE 0.013-0.016 per cell.

Calibration. The four trajectory families are matched at a single anchor point ($t = 5$ weeks): each family's response level at $t = 5$ is set equal to the Gompertz reference level at $t = 5$ via `uniroot` (logistic, hyperbolic-tangent) or closed form (piecewise-linear breakpoint). The asymptotic maximum is held constant across families. Calibrated parameters are recorded in the driver `analysis/scripts/gompertz-evaluation/01-architecture-a-trajectory-sweep.R` and read back by the manuscript chunks for transparency.

6 Results

The simulation programme was executed at 1{,}000 replicates per cell across the 16-cell trajectory-family-by-architecture-by- effect-size factorial in 381 s using 8-core `parallel::mclapply`. Convergence was 100% across all 16{,}000 fits. Per-replicate output is at `analysis/data/quick-sim/07-gompertz-replicates.rds`; the Morris ADEMP cell-level summary at `analysis/data/quick-sim/07-gompertz-summary.txt`. Cell MCSE on power runs 0.013 to 0.016 across the alternative cells and 0.005 to 0.006 on the conservative null cells.

6.1 Power across response-curve families

Trajectory-family power at $c_{bm} = 0.45$ under both architectures, percent $\pm$ MCSE:

Family	Architecture A	Architecture B
Gompertz	76.1 $\pm$ 1.4	78.1 $\pm$ 1.3
Logistic	77.1 $\pm$ 1.3	80.6 $\pm$ 1.3
Hyperbolic tangent	76.9 $\pm$ 1.3	78.0 $\pm$ 1.3
Piecewise-linear breakpoint	77.8 $\pm$ 1.3	77.9 $\pm$ 1.3

We note that under Architecture A (mean moderation), the four families cluster

7 Discussion

7.1 A negative finding, cleanly resolved

We report what is, in essence, a negative finding: at the prazosin-calibrated reference cell (Hybrid OL+BDC, $N = 35$, $c_{bm} = 0.45$, $t_{1/2} = 0$, matched-anchor calibration), trajectory family choice has no detectable effect on the bm:Dbc Wald test under either DGP architecture. The four families span the qualitative range of plausible templates for monotone-saturating treatment-response trajectories: asymmetric (Gompertz), symmetric smooth (logistic and hyperbolic-tangent), and non-smooth (piecewise-linear breakpoint). All four produce statistically indistinguishable power and Type I error at MCSE 0.013 to 0.016 per cell.

The pre-registered hypothesis that the piecewise-linear breakpoint family, with its non-differentiable feature, would behave qualitatively differently from the smooth families is not supported. The breakpoint feature is invisible to the linear-mixed Wald test at the explored sample size and effect size. This is a methodologically valuable null result: if the breakpoint feature were detectable through power loss, the manuscript would have reported a meaningful sensitivity to non-smoothness. Its absence implies that smoothness assumptions in the trajectory family carry no inferential cost in the regime evaluated.

7.2 Why trajectory family does not matter at this resolution

We see three mechanisms that, taken together, account for the negative finding.

Calibration absorbs the differences. The four families are matched at the asymptotic maximum and at a single anchor point ($t = 5$ weeks). At matched marginal effect size, the within-trajectory variation in shape is small relative to the between-replicate variation in the bm:Dbc estimator. The calibration constraint is intentional in the simulation design: it isolates trajectory-shape effects from amplitude effects.

The bm:Dbc Wald test averages over time. The continuous D_{bc} predictor multiplied by the biomarker integrates the trajectory's contribution across all on-drug timepoints, with weights determined by the predictor's exponential-decay structure. Local trajectory-shape differences (where one family rises faster than another between timepoints) are averaged within the integration. The Wald test is therefore relatively insensitive to local-shape features.

Architecture B's covariance moderation is upstream of trajectory shape. Under MVN differential correlation, the biomarker-by-treatment information lives in the $\partial \text{Cor}(BM, BR) / \partial D_{bc}$ structure, which is independent of the trajectory family that BR follows. Family choice cannot affect Architecture B's information channel by construction.

The combined effect is that trajectory family choice is, at this resolution, an inferentially inert axis of the simulation design, and the framework's modGompertz

359 default is in this sense robust by construction. We turn next to the conditions
360 under which that robustness would be expected to break down.

361 7.3 Conditions that would change the answer

362 Four conditions, none exercised in the present 16-cell factorial, would plausibly
363 produce trajectory-family-dependent power.

- 364 1. **Smaller sample size.** At $N = 35$ the bm:Dbc test integrates over a
365 moderately large within-subject sample; smaller N might amplify the local-
366 shape sensitivity.
- 367 2. **Larger or smaller c_{bm} .** The $c_{bm} = 0.45$ reference cell sits in a power
368 regime where families have comparable signal-to-noise. Pushing c_{bm} to 0.30
369 (where power lies near 0.5) might surface family differences via different
370 finite-sample sampling distributions.
- 371 3. **Carryover induces correlation decay.** With $t_{1/2} > 0$, the bm:Dbc-
372 mediated information channel is eroded differently for different trajectory
373 shapes (the integration weights interact with the off-drug exponential decay).
374 The $t_{1/2} = 0$ slice exercised here suppresses this interaction.
- 375 4. **Non-monotone or biphasic shape envelopes.** The four families share
376 a monotone-saturating envelope. Families that peak and decay (biphasic)
377 or oscillate would produce genuinely different trajectories with different
378 bm:Dbc information content.

379 Each condition is the natural axis of a follow-up sensitivity analysis. The 144-
380 cell grid anticipated in the abstract ($\$N \times c_{\{bm\}} \times t_{\{1/2\}} \times \$$ family) is the
381 canonical follow-up, and we recommend it as the production-grade extension that
382 would close out the trajectory-family question at a level of rigour adequate for the
383 methodological literature.

384 7.4 The prazosin/PTSD reference cell

385 For the motivating prazosin-PTSD application that drives the pmsimstats
386 research programme, the present result implies that sensitivity to the modGomp-
387 pertz assumption is small relative to sensitivity to the carryover specification
388 (the subject of the companion manuscript at [analysis/report/02-carryover-](#)
389 [sensitivity/](#)) and to the DGP architecture (the subject of [analysis/report/01-dgp-mean-moderation-vs-m](#)
390 A trial designer using the pmsimstats framework can defend the modGompertz
391 choice without reservation at the reference parameters, and should redirect
392 sensitivity-analysis attention to the carryover and architecture axes that have
393 demonstrably larger inferential consequences.

394 8 Conclusions

395 The modified Gompertz function is a defensible parametric template for
396 monotone-saturating treatment-response trajectories in N-of-1 simulation. At the
397 prazosin/PTSD reference cell with matched-anchor calibration, the four-family

comparison (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint) produces statistically indistinguishable power and Type I error under both DGP architectures at MCSE 0.013 to 0.016 per cell. The pre-registered hypothesis that the piecewise-linear breakpoint family would behave qualitatively differently is not supported.

The negative finding is methodologically informative: it implies that the `bm:Dbc` Wald test is robust to the principal qualitative axes of trajectory-family choice in the regime evaluated. Trial designers using the `pmsimstats` framework can accept the `modGompertz` default without reservation at the reference parameters and direct sensitivity-analysis attention to the carryover and DGP-architecture axes that have demonstrably larger inferential consequences.

The 144-cell extended grid (sample sizes, effect sizes, DGP-side carryover) anticipated in the original abstract is the natural follow-up. The present focused 16-cell factorial addresses the central scientific question with statistical clarity; the extended grid would map the boundary conditions at which trajectory-family choice begins to matter.

9 Conflict of interest

The authors declare no conflicts of interest.

10 Funding

This work received no specific funding.

11 Data availability statement

The data and code that support the findings of this study are openly available in the `pmsimstats-ng` project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

12 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's `zzcollab` Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. Each simulation driver sets `set.seed(20260507)`

433 at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`.
434 Per-replicate seeds derive from the master seed by `+ rep_idx`.

435 **Data and code availability.** Per-replicate simulation outputs are checkpointed
436 at `analysis/data/quick-sim/07-gompertz-replicates.rds`; Morris ADEMP
437 cell-level summaries at the companion `07-gompertz-summary.txt`. Driver scripts
438 are under `analysis/scripts/gompertz-evaluation/`. The pre-registered
439 ADEMP plan is at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.md`.
440 The manuscript source, paper-specific bibliography (`references.bib`), and SIM
441 CSL file are at `analysis/report/07-gompertz-evaluation/`.

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